

Alexis Brissard

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ACADEMIA

Mila - Québec AI Institute

Jun. 2026 – Present

Collaborating Researcher

Montréal, Québec

- **Research work:** Perform a Life Cycle Analysis of Mila's computing infrastructure to attribute environmental impacts to research domains and usage types. Redaction of an internal report with mitigation strategies and an academic paper to share findings to the broader scientific community.

École Polytechnique de Montréal

Aug. 2023 – Apr. 2026

Research M.Sc. in Computer Science, With Distinction, GPA: 4.00 / 4.00

Montréal, Québec

- **Research work:** Designed a fine-grained evaluation framework for LLM-based process modeling using information extraction and semantic matching techniques. Created 2 datasets and benchmarked textual representations to address standardization and data scarcity challenges in a low-resource domain.
- **Selected course work:**
 - * Representation Learning | Grade: A | *Mila*
 - * Introduction to AI Ethics | Grade: A | *McGill University*
 - * Sustainable Degrowth: Theories and Practices | Audited | *HEC Montréal*

Institut National des Sciences Appliquées de Lyon

Aug. 2018 – Aug. 2023

Engineering Degree in Computer Science (B.Eng. + M.Eng.)

Lyon, France

TEACHING

Trustworthy and Responsible AI Learning (TRAIL): Sustainability module | *Instructor*

Autumn 2026

INF8460 - Natural Language Processing | *Teaching Assistant*

Autumn 2025

INF8225 - Artificial Intelligence: Probabilistic and Learning Techniques | *Teaching Assistant*

Winter 2025

PUBLICATIONS

1. C. Kowalczyk, **A. Brissard**, F. Cuppens, and A. Zouaq, "PETALE: a Fine-grained Evaluation Framework," *Accepted at EM'2026*.
2. **A. Brissard**, F. Cuppens, and A. Zouaq, "Process Modeling with Large Language Models: Towards a Fine-Grained Evaluation Framework," *Master thesis, Polytechnique Montréal, 2026*.
3. **A. Brissard**, F. Cuppens, and A. Zouaq, "What is the Best Process Model Representation? A Comparative Analysis for Process Modeling with Large Language Models," *AI4BPM @ BPM'2025*.

INDUSTRY EXPERIENCE

Chief Technical Officer & Co-founder

Jun. 2025 – Present

Novalytics

Montréal, Québec

- Co-founded a sustainability analytics startup with two co-founders, now incubated at Mila & Finance Montréal
- Built an ESG data pipeline from the ground up, from corporate disclosure extraction to dataset consolidation
- Developing forecasting models to predict ESG performance and likelihood of meeting climate targets

Cybersecurity Consultant

Sep. 2022 – Feb. 2023

Lyvoc

Lyon, France

- Integrated Okta identity management software for 3 clients (5000 to 10000 employee identities)
- Designed a complete architecture for the validation of contractual documents (SharePoint + Vaadin)
- Monitored and presented new technologies (passkeys, new Okta features) to the team (20+ employees)

Web Design and Development Engineer

Jun. 2022 – Aug. 2022

Dassault Systems

Paris, France

PROJECTS

- Mila Hackathon: Safer AI for Youth Mental Health** | *Python, NLP, LLMs* Mar. 2026
- Built SAFEHELP-CAN, a 10k-dialogue dataset to model youth crisis scenarios for the Kids Help Phone hotline.
 - Deployed a BM25+MLP guardrail with < 1ms latency, a 4,000× speedup over LLM-as-judge baselines for triage.
 - Systematized red-teaming with GPT-OSS to identify and mitigate critical failures in the Kids Help Phone chatbot.
- JustitIA** | *Python, Flask, React, NextJS, Pinecone, OpenAI API* Apr. 2024
- Developed a full-stack RAG application to facilitate access to companies legal documents (T&C, Privacy Policy)
 - Set up a Pinecone vector database to store document embeddings
 - Implemented a Flask REST API with React + NextJS as the frontend

SELECTED SERVICE

- Mila Student Association** Mar. 2026 – Present
- Helped create the Mila student union as a member of the executive team
- Mila's Sustainability and Social Committees** Oct. 2024 – Present
- Organized events to raise awareness about climate change and bring the community together
- Harms and Risks of AI in the Military Workshop** Dec. 2024

CERTIFICATES

- Trustworthy and Responsible AI Learning (TRAIL)** | *Mila* Oct. 2024
- Okta Administrator & Okta Developer** Jan. 2023

LANGUAGES

- French** | *Native speaker*
English | *Fully proficient*
German, Japanese | *Notions*

TECHNICAL SKILLS

- Research:** Python, L^AT_EX, Zotero
Web dev: Typescript, Next.js, React, Tailwind, shadcn/ui, AWS, Supabase, Flask
Machine Learning: PyTorch, TensorFlow, scikit-learn, Pinecone, Wandb
LLMs: Hugging Face, vLLM, Google Vertex AI, OpenAI API, Azure Foundry
Dev tools: Git, uv, SLURM, agentic coding

INTERESTS

- Graphic Design** | *Data visualization and typography*
Biking | *Moving around and repairing*
Climbing & Bouldering | *Indoors and outdoors*
Handball | *Goalkeeper at university level and 4 years in a club*
Chess | *Two-time finalist in the French school championships*

ABOUT ME

I'm a Collaborating Researcher at Mila, working on the environmental impacts of data centers under the supervision of Prof. Alex Hernandez-Garcia and Maude Gohier-Murray. My current work applies Life Cycle Analysis to Mila's computing infrastructure, attributing environmental impacts to research domains and usage types.

As an engineer and researcher, I'm concerned about the impact of technologies on our societies and ecosystems. I used the knowledge acquired during my M.Sc. on information extraction to co-found Novalytics, a startup where I build sustainability analytics that help investors cut through greenwashing. That same concern also pushed me to switch academic research direction, moving toward analyzing AI's environmental footprint, focusing on the data centers that power it.

I work to raise environmental awareness both internally at Mila, through its Sustainability Committee, and externally with grassroots climate activism groups in Montréal.